DEVIN THOMAS

PHD CANDIDATE • UNIVERSITY OF NEW HAMPSHIRE

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EDUCATION

UNIVERSITY OF NEW HAMPSHIRE
PhD in Computer Science 2018-Dec 2026 (Expected)
Dissertation topic: Planning in a Dynamic World, Advisor: Dr. Wheeler Ruml

UNIVERSITY OF BRITISH COLUMBIA
BSc with Combined Honours in Physics and Astronomy 2012-2016
Thesis topic: Detecting Point Sources With CHIME Pathfinder, Advisor: Dr. Gary Hinshaw

AWARDS

Best Paper Award, International Symposium on Combinatorial Search 2024
For Thomas, Ruml, and Shimony (2024), "Real-time Safe Interval Path Planning".

JOURNAL PUBLICATIONS

Jordan G. Kueneman, Ernesto Bonadies, Devin Wild Thomas, David W. Roubik, and William T. Wcislo (2023), "Neotropical bee microbiomes point to a fragmented social core and strong species-level effects", *Microbiome*.

Thomas H. Hampton, Devin Wild Thomas, Christopher van der Gast, George A. O'Toole, and Bruce A. Stanton (2021), "Mild cystic fibrosis lung disease is associated with bacterial community stability", *Microbiology Spectrum*.

Laura C. Crane, Jason S. Goldstein, Devin Wild Thomas, Kayla S. Rexroth, and Alison W. Watts (2021), "Effects of life stage on eDNA detection of the invasive European green crab (Carcinus maenas) in estuarine systems", *Ecological Indicators*.

CONFERENCE PUBLICATIONS

Devin Wild Thomas and Wheeler Ruml (2026a), "Complete Real-Time Planning in a Dynamic World", *Under Review*.

Devin Wild Thomas, Ilya Atkin, and Thomas Butkiewicz (2026), "GapFiller 2.0: Web-based Planning Tools for Ocean Mapping in Support of Seabed 2030 Efforts", *OCEANS*.

Devin Wild Thomas and Wheeler Ruml (2026b), "Learning, Commitment, and Completeness in Real-Time Heuristic Search", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*.

Issa Hanou, Eric Kemmeren, Devin Wild Thomas, and Mathijs de Weerdt (2026), "Precomputing Multi-Agent Path Replanning Using Temporal Flexibility", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*.

Devin Wild Thomas and Wheeler Ruml (2025), "Real-time Cost-algebraic Heuristic Search", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*.

Devin Wild Thomas, Wheeler Ruml, and Solomon Eyal Shimony (2024), "Real-time Safe Interval Path Planning", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*.

*Issa K. Hanou, *Devin Wild Thomas, Wheeler Ruml, and Mathijs de Weerdt (2024), "Replanning in Advance for Instant Delay Recovery in Multi-Agent Applications: Rerouting Trains in a Railway Hub", *Proceedings of the International Conference on Automated Planning and Scheduling (ICAPS)*.

Devin Wild Thomas, Tianyi Gu, Wheeler Ruml, and Solomon Eyal Shimony (2022), "Situated Grid Pathfinding Among Moving Obstacles (Extended Abstract)", *Proceedings of the International Symposium on Combinatorial Search (SoCS)*.

WORKSHOP PUBLICATIONS

Devin Wild Thomas, Stephen Wissow, Michael Bauer, Paige McAfee, and Wheeler Ruml (2026), "Bounded-suboptimal Beam Search", *ICAPS 2026 Heuristics and Search for Domain-Independent Planning Workshop (HSDIP)*.

Devin Wild Thomas, Solomon Eyal Shimony, Wheeler Ruml, Erez Karpas, Shahaf S. Shperberg, and Andrew Coles (2023), "Any-Start-Time Planning for SIPP", *ICAPS 2023 Heuristics and Search for Domain-Independent Planning Workshop (HSDIP)*.

UNREFEREED PUBLICATIONS

Devin Wild Thomas, Solomon Eyal Shimony, Wheeler Ruml, Erez Karpas, Shahaf S. Shperberg, and Andrew Coles (2026), *Optimal Planning in a Dynamic World*, arXiv: `2610.03312 [cs.AI]`, (currently in preparation for journal submission).

Alison Watts, Laura Crane, Jason Garwood, Jason Goldstein, Megan Lamb, Christopher Peter, Yoshimi Rii, Shon Schooler, Devin Wild Thomas, William Thomas, et al. (2022), "Red Fish, Blue Fish, Native Fish, New Fish: eDNA as a Tool to Monitor Fish in Estuarine Systems", *Preprint*.

RESEARCH EXPERIENCE

UNIVERSITY OF NEW HAMPSHIRE

Durham, NH

Research Assistant: 2025-Current
Research Group: Data Vis Lab at the Center for Coastal and Ocean Mapping
Project: Sonar Coverage Planning
Technologies: C++, Python, Coverage Planning, Cartographic Projections
Responsibilities:

Designed and implemented the planners for GapFiller 2.0, a web tool for sonar survey planning to support Seabed 2030.
Gap Filling Mode plans covering transits for polygonal unmapped regions
Transit Mode plans small changes to long transits to maximize new sonar coverage.
Deployed at: gapfiller.unh.edu

Research Assistant: 2022-2025
Research Group: UNH AI Group
Project: Planning in a Dynamic World
Technologies: C++, AI Planning, Heuristic Search, Real-time Search
Responsibilities:

Developed heuristic search-based approaches for planning in dynamic worlds.

Bioinformaticist: 2018-2022
Research Group: School of Marine Science and Ocean Engineering
Project: Aquatic Environmental DNA Detection
Technologies: Python, Bioinformatics, eDNA, Metabarcoding
Responsibilities:

Implemented analysis approaches for eDNA metabarcoding projects. Supported the National Estuarine Research Reserve System to set up early detection for invasive species. Tasks included data analysis, water sampling, and mentoring undergraduate researchers.

Summer Undergraduate Research Assistant: 2012-2014
Research Group: UNH Space Science Center
Project: ICME Analysis
Technologies: Python, Space Science
Responsibilities:

- Studied interplanetary coronal mass ejection data from the ACE satellite.

UNIVERSITY OF BRITISH COLUMBIA
Vancouver, Canada

Undergraduate Research Assistant: 2012-2014 **Research Group:** Moerman Lab
Project: Reagent Prep **Technologies:** Laboratory Methods
Responsibilities:
Prepared reagents and other minor tasks to aid the study of gene knockout in *C. elegans*.

TEACHING EXPERIENCE

UNIVERSITY OF NEW HAMPSHIRE
Durham, NH

Teaching Assistant: 2023-2025 **Department:** Computer Science
Courses:
CS 730/830 — Introduction to Artificial Intelligence
CS 758/858 — Algorithms
CS 515 — Data Structures and Introduction to Algorithms
CS 419 — Computer Science for Engineers and Scientists
CS 415 — Introduction to Computer Science I

Bioinformaticist and Guest Lecturer: 2016-2022 **Research Group:** Hubbard Center for Genome Studies
Project: NH-INBRE Bioinformatics Curriculum **Technologies:** Python, Bioinformatics
Responsibilities:
Helped develop and implement the NH-INBRE bioinformatics curriculum, supporting the introduction of modern 'big-data' bioinformatics to undergraduate students.
Guest Lecturer at Dartmouth, University of New Hampshire, Bates, Colby-Sawyer, Keene State College, New England College, Plymouth State University, Saint Anselm College, and Phillips Exeter Academy.
Workshop Instructor for:
Train the Trainers Workshop 2018-2022 Mount Desert Island Biological Lab
Benthic Invertebrate Taxonomy, Metagenomics, and Bioinformatics 2017-2018 Harte Institute, TAMU-CC

Instructor: 2016 **Department:** Genetics
GEN 712/812 — Programming for Bioinformatics: a hybrid undergraduate/graduate course on using Python programming for genomics research. Co-taught with a Ph.D. student, splitting work 50/50; duties included developing course material, lecturing, and grading.

Project SMART Mentor 2012-2014 **Department:** UNH Space Science Center
- Mentored high school students with their Project SMART research projects.

PHILLIPS EXETER ACADEMY
Exeter, New Hampshire

Long-Term Substitute Instructor: 2023 **Department:** Computer Science
Took over instruction of three sections of two introductory CS courses.

- Successfully resumed instruction after disruption of short-term substitute and uncertain return of original instructor.
- Restarted Harkness discussions on current topics in computer science, such as E-Waste, generative AI, and the ethics of self-driving cars.

PROFESSIONAL ACTIVITIES

Organizing

- ICAPS Workshop on Heuristics and Search for Domain-Independent Planning 2025, 2026
- VP Academic of the UBC Physics Society 2015, 2016

Journal Reviewing

- Artificial Intelligence

Conference Reviewing

- International Conference on Automated Planning and Scheduling
- International Symposium on Combinatorial Search
- Workshop on Graph-Theoretic Concepts in Computer Science

Workshop Reviewing

- Workshop on Heuristics and Search for Domain-Independent Planning

Current as of: October 12, 2026